HW #12 (OPTIONAL)

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Question 1. In this question, we will prove the quartic reciprocit law in an analytic way.

Let $K = \mathbb{Q}(\sqrt{-1})$, embedded as a subfield of $\mathbb{C}$ as $K \hookrightarrow \mathbb{C}$, $\sqrt{-1} \mapsto i$, and we will identify $\sqrt{-1} \in K$ with $i \in \mathbb{C}$. Let $\pi = 1 - i$. Recall that in HW#11, Question 3, we defined that, for a rational prime $p \equiv 1 \pmod 4$, a **primary number** is a generator of a prime ideal dividing (p) that is congruent to $1 \pmod{\pi^3}$. For each prime ideal of $\mathcal{O}_K$ dividing p, there is a unique generator that is a primary number.

Let π_1, π_2 be primary numbers of K dividing rational primes $p_1, p_2 \equiv 1 \pmod 4$, and suppose that $p_1 \neq p_2$. For $j = 1, 2$, we let $\chi_j(n) := \left(\frac{n}{\pi_j}\right)$ as a Dirichlet character mod p_j.

(1) For $j = 1, 2$, show that $G(\chi_j)^4 = p_j J(\chi_j, \chi_j)^2$.

Hint. Show that χ_j^2, when evaluated at integers, is the Legendre symbol mod p_j.

(2) For $j = 1, 2$, show that

$$J(\chi_j, \chi_j) = \begin{cases} \pi_j & \text{if } p_j \equiv 5 \pmod 8, \\ -\pi_j & \text{if } p_j \equiv 1 \pmod 8. \end{cases}$$

Hint. Let $p_j = 2k_j - 1$. Show that $J(\chi_j, \chi_j) \equiv \chi_j(k_j)^2 + 2(k_j - 2) \pmod{\pi^3}$ by using that any $u \in \mathcal{O}_K$ satisfies $u \equiv 1 \pmod \pi$. Show that

$$\chi_j(k_j)^2 = \chi_j(2)^2 = \begin{cases} 1 & \text{if } p_j \equiv 1 \pmod 8, \\ -1 & \text{if } p_j \equiv 5 \pmod 8. \end{cases}.$$

Use Lemma 6.25 to finish the proof.

(3) Note that, for $j = 1, 2$, $\overline{\pi}_j$, the complex conjugate of π_j, is a primary number. Show that

$$\chi_1(p_2) = \chi_2(p_1 \overline{\pi}_1^2).$$

Hint. Mimic the analytic proof of cubic reciprocity law as in the textbook, by computing $G(\psi_1)^{p_2}$ in two different ways, where $\psi_j(n) := \left(\frac{n}{\overline{\pi}_j}\right)$.

(4) Show that

$$\left(\frac{p_2}{\pi_1}\right) = \left(\frac{\pi_1}{p_2}\right) \quad \left(\text{namely, } \left(\frac{\pi_2}{\pi_1}\right)\left(\frac{\overline{\pi}_2}{\pi_1}\right) = \left(\frac{\pi_1}{\pi_2}\right)\left(\frac{\pi_1}{\overline{\pi}_2}\right)\right).$$

Hint. Use that $\overline{\chi_2(\overline{\pi_1})} = \psi_2(\pi_1)$.

(5) Let $q \equiv 3 \pmod 4$ be a rational prime. Show that

$$\left(\frac{-q}{\pi_1}\right) = \left(\frac{\pi_1}{-q}\right).$$

Hint. Compute $G(\chi_1)^{q+1}$ in two different ways. Along the way, it is useful to note that $\overline{\pi} \equiv \pi^q \pmod{q}$.

(6) Combine (4) and (5) to show that, for any integer n with $n \equiv 1 \pmod{4}$ and $(n, p_1) = 1$,

$$\left(\frac{n}{\pi_1}\right) = \left(\frac{\pi_1}{n}\right).$$

(7) For any odd integers $a, b \in \mathbb{Z}$ with $(a, b) = 1$, show that $\left(\frac{a}{b}\right) = 1$.

(8) For $j = 1, 2$, let

$$\widetilde{\pi}_j := \begin{cases} \pi_j & \text{if } \pi_j \equiv 1 \pmod{4}, \\ -\pi_j & \text{if } \pi_j \equiv 3 + 2i \pmod{4}. \end{cases}$$

In particular, $\widetilde{\pi}_j = a_j + b_j i$ with $a_j \equiv 1 \pmod{4}$. Show that

$$\left(\frac{\pi_1}{\pi_2}\right)\left(\frac{\pi_2}{\pi_1}\right)^{-1} = \left(\frac{\widetilde{\pi}_1}{\pi_2}\right)\left(\frac{\widetilde{\pi}_2}{\pi_1}\right)^{-1}.$$

(9) Show that

$$\left(\frac{\widetilde{\pi}_1}{\pi_2}\right)\left(\frac{\widetilde{\pi}_2}{\pi_1}\right)^{-1} = \left(\frac{a_2}{\pi_2}\right)^{-1}\left(\frac{a_1}{\pi_1}\right)\left(\frac{a_1 a_2 + b_1 b_2}{\overline{\pi}_1 \pi_2}\right).$$

Use (6) and (7) to show that

$$\left(\frac{\widetilde{\pi}_1}{\pi_2}\right)\left(\frac{\widetilde{\pi}_2}{\pi_1}\right)^{-1} = \left(\frac{i}{a_2}\right)^{-1}\left(\frac{i}{a_1}\right)\left(\frac{i}{a_1 a_2 + b_1 b_2}\right).$$

(10) If $a \in \mathbb{Z}$ is congruent to $1 \pmod{4}$, show that

$$\left(\frac{i}{a}\right) = (-1)^{\frac{a-1}{4}}.$$

(11) Deduce the quartic reciprocity law,

$$\left(\frac{\pi_1}{\pi_2}\right)\left(\frac{\pi_2}{\pi_1}\right)^{-1} = (-1)^{\frac{N_{K/\mathbb{Q}}(\pi_1)-1}{4}\frac{N_{K/\mathbb{Q}}(\pi_2)-1}{4}}.$$

Proof.

(1) By definition

$$J(\chi_j, \chi_j) = \frac{G(\chi_j)^2}{G(\chi_j^2)}.$$

Thus, the statement will follow if we show that $G(\chi_j^2)^2 = p_j$. On the other hand, for $n \in \mathbb{Z}$, by definition, $\chi_j^2(n) \equiv n^{\frac{p_j-1}{2}} \pmod{\pi_j}$, but $n^{\frac{p_j-1}{2}}$ is congruent to the Legendre symbol mod p_j, so by §6.4 of the text, $G(\chi_j^2) = \sqrt{p_j}$, as desired.

(2) Let $p_j = 2k_j - 1$. Then

$$J(\chi_j, \chi_j) = \sum_{a=2}^{p_j-1} \chi_j(a)\chi_j(1-a) = \chi_j(k_j)^2 + 2\sum_{a=2}^{k_j-1} \chi_j(a)\chi_j(1-a).$$

Note that, as $u \in \mathcal{O}_K$ satisfies $u \equiv 1 \pmod{\pi}$ (both 2 and $1 \pm i$ are divisible by $1-i$), for any $2 \le a \le k_j - 1$, $\chi_j(a)\chi_j(1-a) \equiv 1 \pmod{\pi}$. Thus,

$$J(\chi_j, \chi_j) = \sum_{a=2}^{p_j-1} \chi_j(a)\chi_j(1-a) \equiv \chi_j(k_j)^2 + 2(k_j - 2) \pmod{\pi^3}.$$

On the other hand, $\chi_j(2k_j) = 1$, which means that $\chi_j(k_j)^2 = \chi_j(2)^{-2}$. As $\chi_j(2)^2 \in \{\pm 1\}$, $\chi_j(k_j)^2 = \chi_j(2)^{-2} = \chi_j(2)^2$. This is the same as the Legendre symbol of 2 modulo p_j, so, by quadratic reciprocity, it is 1 if $p_j \equiv 1 \pmod 8$, and -1 if $p_j \equiv 5 \pmod 8$. Note also that, as k_j is odd, $2(k_j - 2) \equiv 2 \pmod 4$, so $\equiv 2 \pmod{\pi^3}$. Therefore,

$$J(\chi_j, \chi_j) \equiv \begin{cases} 3 \pmod{\pi^3} & \text{if } p_j \equiv 1 \pmod 8, \\ 1 \pmod{\pi^3} & \text{if } p_j \equiv 5 \pmod 8. \end{cases}$$

This implies that $J(\chi_j, \chi_j) \equiv 1 \pmod{\pi^3}$ if $p_j \equiv 5 \pmod 8$, and $-J(\chi_j, \chi_j) \equiv 1 \pmod{\pi^3}$ if $p_j \equiv 1 \pmod 8$.

Note finally that $J(\chi_j, \chi_j) \in \mathcal{O}_K$ whose norm is p_j. Therefore, either $J(\chi_j, \chi_j)$ (if $p_j \equiv 5 \pmod 8$) or $-J(\chi_j, \chi_j)$ (if $p_j \equiv 1 \pmod 8$) is a primary number. By Lemma 6.25, it follows that $J(\chi_j, \chi_j) \equiv 0 \pmod{\pi_j}$, which implies the desired statement.

(3) Firstly

$$G(\psi_1)^{p_2} = \left(\sum_{a=1}^{p_1-1} \psi_1(a) e^{\frac{2\pi i a}{p_1}} \right)^{p_2} \equiv \sum_{a=1}^{p_1-1} \psi_1(a)^{p_2} e^{\frac{2\pi i a p_2}{p_1}} = \sum_{a=1}^{p_1-1} \psi_1(a) e^{\frac{2\pi i a p_2}{p_1}}$$

$$= \psi_1(p_2)^{-1} \sum_{a=1}^{p_1-1} \psi_1(p_2 a) e^{p_2 \cdot \frac{2\pi i a}{p_1}} = \psi_1(p_2)^3 G(\psi_1) \pmod{\pi_2}.$$

On the other hand, by (1) and (2), $G(\chi_j)^4 = p_j \pi_j^2$, so applying this to ψ_1 and $\overline{\pi}_1$,

$$G(\psi_1)^{p_2-1} = p_1^{\frac{p_2-1}{4}} \overline{\pi}_1^{\frac{p_2-1}{2}} \equiv \left(\frac{p_1 \overline{\pi}_1^2}{\pi_2} \right) \pmod{\pi_2}.$$

This implies that $\psi_1(p_2)^3 \equiv \chi_2(p_1 \overline{\pi}_1^2) \pmod{\pi_2}$. Since both sides are in μ_4, this implies that $\psi_1(p_2)^3 = \chi_2(p_1 \overline{\pi}_1^2)$. Note also that $\psi_1(p_2)^3 = \overline{\psi_1(p_2)} = \chi_1(p_2)$, so we are done.

(4) By (3), it suffices to show that $\chi_2(p_2 \overline{\pi}_1^2) = \chi_2(\pi_1)\psi_2(\pi_1)$, or $\chi_2(\overline{\pi}_1^3) = \psi_2(\pi_1)$. This is indeed true as $\chi_2(\overline{\pi}_1^3) = \overline{\chi_2(\overline{\pi}_1)} = \psi_2(\pi_1)$.

(5) As $G(\chi_1) = p_1 \pi_1^2$,

$$G(\chi_1)^{q+1} = p_1^{\frac{q+1}{4}} \pi_1^{\frac{q+1}{2}} = \overline{\pi}_1^{\frac{q+1}{4}} \pi_1^{\frac{3q+3}{4}}.$$

Note that, as q is inert in K, $\mathcal{O}_K/(q) \cong \mathbb{F}_{q^2}$, and $D(q|q) = \mathrm{Gal}(K/\mathbb{Q})$, so that $\mathrm{Fr}_q \in D(q|q)$ is the complex conjugation; in particular, $\overline{\pi}_1 \equiv \pi_1^q \pmod q$. Therefore,

$$G(\chi_1)^{q+1} \equiv \pi_1^{\frac{q(q+1)+3q+3}{4}} = \pi_1^{\frac{q^2+4q+3}{4}} = \pi_1^{\frac{q^2-1}{4}} \pi_1^{q+1} \equiv \left(\frac{\pi_1}{q} \right) \pi_1^{q+1} \pmod q.$$

On the other hand,

$$G(\chi_1)^q = \left(\sum_{a=1}^{p_1-1} \chi_1(a) e^{\frac{2\pi i a}{p_1}} \right)^q \equiv \sum_{a=1}^{p_1-1} \chi_1(a)^q e^{\frac{2\pi i a q}{p_1}} = \sum_{a=1}^{p_1-1} \psi_1(a) e^{\frac{2\pi i a q}{p_1}}$$

$$= \psi_1(q)^{-1} \sum_{a=1}^{p_1-1} \psi_1(qa) e^{q \cdot \frac{2\pi i a}{p_1}} = \chi_1(q) G(\psi_1) \pmod q.$$

Therefore,

$$G(\chi_1)^{q+1} \equiv \chi_1(q) G(\chi_1) G(\overline{\chi}_1) = \chi_1(q) G(\chi_1) \chi_1(-1) \overline{G(\chi_1)} = \chi_1(-q) p_1 \pmod q.$$

Note again that $p_1 = \pi_1 \overline{\pi}_1 \equiv \pi_1^{q+1} \pmod q$, so this implies that $\chi_1(-q) \equiv \left(\frac{\pi_1}{q}\right) = \left(\frac{\pi_1}{-q}\right) \pmod q$. As both sides are in μ_4, this is actually an equality, and we get the desired conclusion.

(6) Let $n = \prod_{i=1}^m q_i^{e_i}$ be a prime factorization of n (as integers). Note that, as $n \equiv 1 \pmod 4$, prime numbers $\equiv 3 \pmod 4$ appear exactly even number of times in the prime factorization. Thus, if we define $q_i' = q_i$ if $q_i \equiv 1 \pmod 4$ and $q_i' = -q_i$ if $q_i \equiv 3 \pmod 4$, then $n = \prod_{i=1}^m (q_i')^{e_i}$. To show (6), it suffices to show that $\left(\frac{q_i'}{\pi_1}\right) = \left(\frac{\pi_1}{q_i'}\right)$. If $q_i \equiv 3 \pmod 4$, this is (5). If $q_i \equiv 1 \pmod 4$, then $q_i = \tau\overline{\tau}$ for primary numbers $\tau, \overline{\tau}$, so we can apply (4).

(7) It suffices to prove when b is a prime number.

- If $b \equiv 1 \pmod 4$, then $b = \tau\overline{\tau}$ for primary numbers τ and $\overline{\tau}$, so if $\epsilon \in \{\pm 1\}$ such that $\epsilon a \equiv 1 \pmod 4$, by (6),

$$\left(\frac{\epsilon a}{b}\right) = \left(\frac{\epsilon a}{\tau}\right)\left(\frac{\epsilon a}{\overline{\tau}}\right) = \left(\frac{\tau}{\epsilon a}\right)\left(\frac{\overline{\tau}}{\epsilon a}\right) = \left(\frac{b}{a}\right).$$

  Thus, we only need to prove when $a = -1$. However, then $\left(\frac{-1}{\overline{\tau}}\right) = \overline{\left(\frac{-1}{\tau}\right)} = \left(\frac{-1}{\tau}\right)^{-1}$, so $\left(\frac{-1}{b}\right) = 1$.
- If $b \equiv 3 \pmod 4$, then as b is inert in K,

$$\left(\frac{a}{b}\right) \equiv a^{\frac{b^2-1}{4}} \pmod b.$$

  However, as $a \in \mathbb{Z}$, already $a^{b-1} \equiv 1 \pmod b$, so

$$\left(\frac{a}{b}\right) \equiv (a^{b-1})^{\frac{b+1}{4}} \equiv 1 \pmod b.$$

(8) Note that

$$\left(\frac{-1}{\pi_j}\right) = (-1)^{\frac{p_j-1}{4}} = \begin{cases} 1 & \text{if } p_j \equiv 1 \pmod 8, \\ -1 & \text{if } p_j \equiv 5 \pmod 8. \end{cases}$$

Note also that $\pi_j \equiv 1 \pmod 4$ if and only if $p_j \equiv 1 \pmod 8$. Indeed, if $\pi_j \equiv 1 \pmod 4$, then $\pi_j = 1 + 4a$ for some $a \in \mathbb{Z}[i]$, so $p_j = (1+4a)(1+4\overline{a}) = 1 + 4(a+\overline{a}) + 16a\overline{a} \equiv 1 \pmod 8$ as $a + \overline{a}$ is always an even integer. On the other hand, if $p_j \equiv 1 \pmod 8$, then $p_j = x^2 + y^2$ for integers x, y such that y is a multiple of 4 and x is odd. By possibly taking $-x$ instead of x, we may assume that $x \equiv 1 \pmod 4$. Then $x \pm iy \equiv 1 \pmod 4$ are the two primary numbers dividing p_j, so $\pi_j \equiv 1 \pmod 4$. Therefore, there are four cases:

- If $p_1 \equiv p_2 \equiv 1 \pmod 8$, $\widetilde{\pi}_j = \pi_j$, and there is nothing to prove.
- If $p_1 \equiv 1 \pmod 8$ and $p_2 \equiv 5 \pmod 8$, $\widetilde{\pi}_1 = \pi_1$ and $\widetilde{\pi}_2 = -\pi_2$, and we need to show that $\left(\frac{-1}{\pi_1}\right) = 1$, which is indeed true.
- If $p_1 \equiv 5 \pmod 8$ and $p_2 \equiv 1 \pmod 8$, this is the same (with the roles of p_1, p_2 switched) as the above case.

- If $p_1 \equiv p_2 \equiv 5 \pmod 8$, $\widetilde{\pi}_j = -\pi_j$, and we need to show that $\left(\frac{-1}{p_1}\right) = \left(\frac{-1}{p_2}\right)$, which is true as both sides are -1.

(9) Note that $b_j \equiv a_j i \pmod{\pi_j}$. Thus,

$$\left(\frac{a_2\widetilde{\pi}_1}{\pi_2}\right) = \left(\frac{a_1a_2 + b_1a_2i}{\pi_2}\right) = \left(\frac{a_1a_2 + b_1b_2}{\pi_2}\right),$$

$$\left(\frac{a_1\widetilde{\pi}_2}{\pi_1}\right) = \left(\frac{a_1a_2 + a_1b_2i}{\pi_1}\right) = \left(\frac{a_1a_2 + b_1b_2}{\pi_1}\right).$$

Thus

$$\left(\frac{\widetilde{\pi}_1}{\pi_2}\right)\left(\frac{\widetilde{\pi}_2}{\pi_1}\right)^{-1} = \left(\frac{a_2}{\pi_2}\right)^{-1}\left(\frac{a_1}{\pi_1}\right)\left(\frac{a_1a_2 + b_1b_2}{\pi_2}\right)\left(\frac{a_1a_2 + b_1b_2}{\pi_1}\right)^{-1} = \left(\frac{a_2}{\pi_2}\right)^{-1}\left(\frac{a_1}{\pi_1}\right)\left(\frac{a_1a_2 + b_1b_2}{\overline{\pi}_1\pi_2}\right),$$

as $\overline{a_1a_2 + b_1b_2} = a_1a_2 + b_1b_2$. By (6),

$$\left(\frac{\widetilde{\pi}_1}{\pi_2}\right)\left(\frac{\widetilde{\pi}_2}{\pi_1}\right)^{-1} = \left(\frac{\pi_2}{a_2}\right)^{-1}\left(\frac{\pi_1}{a_1}\right)\left(\frac{\overline{\pi}_1\pi_2}{a_1a_2 + b_1b_2}\right) = \left(\frac{b_2i}{a_2}\right)^{-1}\left(\frac{b_1i}{a_1}\right)\left(\frac{\overline{\pi}_1\pi_2}{a_1a_2 + b_1b_2}\right).$$

By (7),

$$\left(\frac{\widetilde{\pi}_1}{\pi_2}\right)\left(\frac{\widetilde{\pi}_2}{\pi_1}\right)^{-1} = \left(\frac{i}{a_2}\right)^{-1}\left(\frac{i}{a_1}\right)\left(\frac{\overline{\pi}_1\pi_2}{a_1a_2 + b_1b_2}\right).$$

Note also that $\overline{\pi}_1\pi_2 = (a_1 - b_1i)(a_2 + b_2i) = (a_1a_2 + b_1b_2) + (a_1b_2 - b_1a_2)i$, so by (6) and (7) again,

$$\left(\frac{\widetilde{\pi}_1}{\pi_2}\right)\left(\frac{\widetilde{\pi}_2}{\pi_1}\right)^{-1} = \left(\frac{i}{a_2}\right)^{-1}\left(\frac{i}{a_1}\right)\left(\frac{(a_1b_2 - b_1a_2)i}{a_1a_2 + b_1b_2}\right) = \left(\frac{i}{a_2}\right)^{-1}\left(\frac{i}{a_1}\right)\left(\frac{i}{a_1a_2 + b_1b_2}\right).$$

(10) We claim that we only need to check when a is a prime $1 \pmod 4$ or $a = -q$ for a prime $q \equiv 3 \pmod 4$. For this, we need to check that, if $a \equiv b \equiv 1 \pmod 4$,

$$(-1)^{\frac{a-1}{4} + \frac{b-1}{4}} = (-1)^{\frac{ab-1}{4}},$$

which is the same as $a + b - 2 \equiv ab - 1 \pmod 8$, or $(a-1)(b-1) \equiv 0 \pmod 8$, which is indeed true.

- If $a \equiv 1 \pmod 4$ is a prime, then $a = \tau\overline{\tau}$ for primary numbers $\tau, \overline{\tau}$. Now by definition

$$\left(\frac{i}{\tau}\right) = \left(\frac{i}{\overline{\tau}}\right) = i^{\frac{a-1}{4}},$$

so we get the result.

- If $a = -q$ for a prime $q \equiv 3 \pmod 4$, then

$$\left(\frac{i}{a}\right) = i^{\frac{q^2-1}{4}}.$$

Thus we want to show that $\frac{q^2-1}{4} \equiv \frac{1+q}{2} \pmod 4$, or $q^2 - 1 \equiv 2 + 2q \pmod{16}$, or $q^2 - 2q - 3 \equiv 0 \pmod{16}$, but $q^2 - 2q - 3 = (q-3)(q+1)$, so both $q - 3$ and $q + 1$ are divisible by 4, so we are done.

(11) By (8), (9) and (10),

$$\left(\frac{\pi_1}{\pi_2}\right)\left(\frac{\pi_2}{\pi_1}\right)^{-1} = (-1)^{-\frac{a_2-1}{4}+\frac{a_1-1}{4}+\frac{a_1a_2+b_1b_2-1}{4}} = (-1)^{\frac{(a_1-1)(a_2+1)+b_1b_2}{4}} = (-1)^{\frac{b_1b_2}{4}},$$

as $a_1 \equiv a_2 \equiv 1 \pmod 4$ so that $(a_1-1)(a_2+1) \equiv 0 \pmod 8$. Note that, for $j = 1, 2$, b_j is even, and $b_j \equiv 2 \pmod 4$ if and only if $p_j \equiv 5 \pmod 8$, so $\left(\frac{\pi_1}{\pi_2}\right)\left(\frac{\pi_2}{\pi_1}\right)^{-1}$ is -1 if and only if $p_1 \equiv p_2 \equiv 5 \pmod 8$ and 1 otherwise, which is exactly the content of the quartic reciprocity law!

□

Question 2. Use various methods we have learned in class to show that $h_{\mathbb{Q}(\sqrt{-31})} = 3$ in three different ways.

Proof. Let $K = \mathbb{Q}(\sqrt{-31})$.

- **First proof: Minkowski bound.** Note that $\operatorname{disc}(K) = -31$. Therefore, the Minkowski bound is
$$B_K = \frac{2}{\pi}\sqrt{31} \sim 3.544\ldots.$$
Therefore, Cl_K is represented by ideals of $\mathcal{O}_K$ of norms $2, 3$. We are thus led to study the splitting of rational primes $2, 3$. Note that $\mathcal{O}_K = \mathbb{Z}[\frac{1+\sqrt{-31}}{2}]$, so $p \in \mathbb{Z}$ splits as $X^2 - X + 8$ splits mod p.
  - If $p = 2$, then $X^2 - X + 8 \equiv X(X-1) \pmod 2$, so $(2) = (2, \frac{1+\sqrt{-31}}{2})(2, \frac{1-\sqrt{-31}}{2})$. Let $\mathfrak{p}_2 = (2, \frac{1+\sqrt{-31}}{2})$, so that $(2) = \mathfrak{p}_2\overline{\mathfrak{p}}_2$ (here $\bar{\cdot}$ is the complex conjugate, as usual).
  - If $p = 3$, then $X^2 - X + 8$ is irreducible mod 3, so 3 is inert.

  In particular, this implies that $h_K \leq 3$. More precisely, $h_K = 3$ if $[\mathfrak{p}_2] \neq 1$ (in this case, $\mathrm{Cl}_K = \{1, [\mathfrak{p}_2], [\overline{\mathfrak{p}}_2]\}$), and $h_K = 1$ if $[\mathfrak{p}_2] = 1$. However, $\mathfrak{p}_2$ is principal if and only if there is a norm 2 element $\alpha \in \mathcal{O}_K$, which is impossible as this means there exist $x, y \in \mathbb{Z}$ such that $x^2 + 31y^2 = \pm 8$, which is impossible.
- **Second proof: Binary quadratic forms.** We can directly apply Theorem 3.29. Note that $d = 31$, so $1 \leq a \leq 3$.
  - For $a = 1$, we check whether, for any $-1 < b \leq 1$, $c = \frac{31+b^2}{4}$ is an integer. This is only possible when b is odd, so $b = 1$. In that case, $c = 8$. Note that $c > a$, so we found a new class of binary quadratic forms.
  - For $a = 2$, we check whether, for any $-2 < b \leq 2$, $c = \frac{31+b^2}{8}$ is an integer. This is only possible when b is odd, so $b = 1$ or -1. In both cases, $c = 8$, and as $c > a$, we found two new classes of binary quadratic forms.
  - For $a = 3$, we check whether, for any $-3 < b \leq 3$, $c = \frac{31+b^2}{12}$ is an integer. However, the numerator is never divisible by 3, as it is congruent to $1 + b^2 \pmod 3$. Therefore, there are no more new classes of binary quadratic forms.

  The above analysis implies that $h_K = 3$.
- **Third proof: Analytic class number formula.** We can directly apply Theorem 6.49(1). Note that $\#\mu_K = 2$, so
$$h_K = \frac{1}{31}\left|\sum_{a=1}^{31} \chi_{-31}(a)a\right|.$$
By quadratic reciprocity, χ_{-31} is the Legendre symbol mod 31.

a	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$\chi_{-31}(a)$	1	-1	1	1	-1	1	1	1	1	-1	-1	-1	1	-1

Thus,

$$\begin{aligned}31h_K &= |(1-30)+(2-29)-(3-28)+(4-27)+(5-26)-(6-25)+(7-24)\\ &+(8-23)+(9-22)+(10-21)-(11-20)-(12-19)-(13-18)+(14-17)-(15-16)|\\ &= |-29-27+25-23-21+19-17-15-13-11+9+7+5-3+1|\\ &= 93,\end{aligned}$$

so $h_K = 3$.

□

Question 3. Let $K/\mathbb{Q}$ be a cubic Galois field. Note that this forces K to only have real embeddings, and therefore $\operatorname{rank}\mathcal{O}_K^\times = 2$. Note that $\operatorname{disc}(K)$ is a perfect square by Question 1 of Midterm.

(1) Let $\operatorname{disc}(K) = f^2$ for $f > 0$. Use the Conductor-discriminant formula, Theorem 7.5, to show that $K \subset \mathbb{Q}(\zeta_f)$.

Hint. Note that, if χ is a nontrivial character of $\operatorname{Gal}(K/\mathbb{Q})$, then the characters of $\operatorname{Gal}(K/\mathbb{Q})$ are $\{\mathbf{1}, \chi, \chi^{-1}\}$. Thus, if m is the minimal integer such that $K \subset \mathbb{Q}(\zeta_m)$ (which exists by the Kronecker–Weber theorem), show that the Conductor-discriminant formula implies that $\operatorname{disc}(K) = m^2$.

(2) Let χ be a nontrivial Dirichlet character of conductor f and order 3 (i.e., $\chi^3 = \mathbf{1}$, $\chi \neq \mathbf{1}$). Show that

$$(**)\qquad h_K R_K = \left|\sum_{a=1}^{\lfloor \frac{f}{2}\rfloor} \chi(a)\log\left(\sin\left(\frac{\pi a}{f}\right)\right)\right|^2.$$

Hint. Use analytic class number formula.

(3) Now consider $K = \mathbb{Q}(\zeta_7)^+$ (cf. Exercise 2.5). Recall that $\frac{\zeta_7^a - 1}{\zeta_7 - 1}$, $1 \le a \le 6$, is a unit in $\mathbb{Q}(\zeta_7)$ (**cyclotomic units**). Show that

$$\epsilon_1 := \zeta_7^3 \frac{\zeta_7^2 - 1}{\zeta_7 - 1}, \quad \epsilon_2 := \zeta_7^{-1}\frac{\zeta_7^3 - 1}{\zeta_7 - 1},$$

are units in K.

(4) Let $\sigma_1 : K \hookrightarrow \mathbb{R}$ be defined as $\zeta_7 + \zeta_7^{-1} \mapsto 2\cos(2\pi/7)$, and $\sigma_2 : K \hookrightarrow \mathbb{R}$ be defined as $\zeta_7 + \zeta_7^{-1} \mapsto 2\cos(6\pi/7)$. Show that

$$\left|\det\begin{pmatrix}\log(|\sigma_1(\epsilon_1)|) & \log(|\sigma_1(\epsilon_2)|)\\ \log(|\sigma_2(\epsilon_1)|) & \log(|\sigma_2(\epsilon_2)|)\end{pmatrix}\right| \sim 0.52545\ldots.$$

Deduce that $\langle \epsilon_1, \epsilon_2\rangle$ generate a finite index subgroup of $\mathcal{O}_K^\times$.

(5) Compute the right hand side of (**) when $K = \mathbb{Q}(\zeta_7)^+$, and show that

$$\text{RHS of (**), when } K = \mathbb{Q}(\zeta_7)^+ \sim 0.52545\ldots.$$

(6) Show that $h_K = 1$.

(7) Let C^+ be the group (multiplicatively) generated by ± 1, ϵ_1, ϵ_2. Show that[1] $\mathcal{O}_K^\times = C^+$.

Proof. (1) Let m be the minimal integer such that $K \subset \mathbb{Q}(\zeta_m)$. Let χ be a nontrivial character of $\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$ trivial on $\mathrm{Gal}(\mathbb{Q}(\zeta_m)/K)$. Then, the other such nontrivial character is χ^{-1}, and both characters have the same conductors. Thus, by the Conductor-discriminant formula, f is the conductor of χ. This implies that $f|m$. By the definition of the conductor, both χ, χ^{-1} are trivial on $\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}(\zeta_f))$, which means that $\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}(\zeta_f)) \subset \mathrm{Gal}(\mathbb{Q}(\zeta_m)/K)$, or $\mathbb{Q}(\zeta_f) \supset K$. This means that $f \geq m$, so $f = m$, as desired.

(2) By analytic class number formula (or Corllary 6.43 which incorporates the factorization of the Dedekind zeta function), we have

$$\frac{4R_K h_K}{f} = \frac{8R_K h_K}{2f} = L(1,\chi)L(1,\chi^{-1}).$$

As $\chi^{-1} = \overline{\chi}$, by Theorem 6.44, $|L(1,\chi)| = |L(1,\chi^{-1})|$, and as $\frac{4R_K h_K}{f}$ is a positive real number, it follows that $L(1,\chi^{-1}) = \overline{L(1,\chi)}$. Therefore, we have

$$\frac{4R_K h_K}{f} = |L(1,\overline{\chi})|^2.$$

Note that χ is even, as $\chi(-1)$ is simultaneously a second root of unity and a third root of unity. Thus, by Theorem 6.44,

$$|L(1,\overline{\chi})| = \frac{1}{\sqrt{f}} \left| \sum_{a=1}^{f} \chi(a) \log \left| 1 - e^{\frac{2\pi i a}{f}} \right| \right|.$$

As χ is even,

$$\left| \sum_{a=1}^{f} \chi(a) \log \left| 1 - e^{\frac{2\pi i a}{f}} \right| \right| = 2 \left| \sum_{a=1}^{\lfloor \frac{f}{2} \rfloor} \chi(a) \log \left| 1 - e^{\frac{2\pi i a}{f}} \right| \right|,$$

noting that $\chi(f/2) = 0$ if f is even (as $(f/2, f) = f/2 > 1$ unless $f = 2$, but $f \neq 4$ as $\mathbb{Q}(\zeta_4)$ cannot contain a cubic subfield). As done in the proof of Theorem 6.49, we have

$$\log \left| 1 - e^{\frac{2\pi i a}{f}} \right| = \log \left(2 \sin \left(\frac{\pi a}{f} \right) \right) = \log 2 + \log \left(\sin \left(\frac{\pi a}{f} \right) \right),$$

so (**) follows as $\sum_{a=1}^{f} \chi(a) = 0$.

(3) Both ϵ_1, ϵ_2 are units in $\mathbb{Q}(\zeta_7)$, so it suffices to show that they are fixed under the Galois conjugation $\zeta_7 \mapsto \zeta_7^{-1}$. Note that

$$\epsilon_1 = \zeta_7^3(\zeta_7 + 1) = \zeta_7^3 + \zeta_7^{-3},$$
$$\epsilon_2 = \zeta_7^{-1}(\zeta_7^2 + \zeta_7 + 1) = \zeta_7 + \zeta_7^{-1} + 1,$$

so clearly they are in K.

(4) The 2×2 matrix is

$$\begin{pmatrix} \log|2\cos(6\pi/7)| & \log|2\cos(2\pi/7)+1| \\ \log|2\cos(4\pi/7)| & \log|2\cos(6\pi/7)+1| \end{pmatrix} = \begin{pmatrix} \log(2\cos(\pi/7)) & \log(2\cos(2\pi/7)+1) \\ \log(2\cos(3\pi/7)) & \log(2\cos(\pi/7)-1) \end{pmatrix}$$

[1] cf. Theorem 7.16

We see that, if we put its determinant into calculator, we get

$$\log(2\cos(\pi/7))\log(2\cos(\pi/7)-1)-\log(2\cos(2\pi/7)+1)\log(2\cos(3\pi/7))\sim 0.52545\ldots.$$

This implies that there is no multiplicative relation between ϵ_1,ϵ_2 (otherwise, this means $\epsilon_1^a=\epsilon_2^b$, which would make the 2×2 matrix singular). As we already know $\operatorname{rank}\mathcal{O}_K^\times=2$, we see that $\langle\epsilon_1,\epsilon_2\rangle\subset\mathcal{O}_K^\times$ is a finite index subgroup.

(5) As 3 is a primitive root mod 7, we may choose χ so that $\chi(3)=e^{2\pi i/3}$ (which implies that $\chi(2)=e^{4\pi i/3}$). Then the RHS of (**) is

$$|\chi(1)\log(\sin(\pi/7))+\chi(2)\log(\sin(2\pi/7))+\chi(3)\log(\sin(3\pi/7))|^2$$

$$=\left(\log(\sin(\pi/7))-\frac{\log(\sin(2\pi/7))-\log(\sin(3\pi/7))}{2}\right)^2+\frac{3}{4}(\log(\sin(2\pi/7))-\log(\sin(3\pi/7)))^2.$$

If you put this into a calculator, you get $\sim 0.52545\ldots$.

(6) Note that $\operatorname{disc}(K)=49$, so the Minkowski bound is

$$B_K=\frac{6}{27}\sqrt{49}=\frac{42}{27}\sim 1.\ldots.$$

Therefore, this alone implies that $h_K=1$.

(7) By (6),

$$R_K=\left|\sum_{a=1}^{\lfloor\frac{f}{2}\rfloor}\chi(a)\log\left(\sin\left(\frac{\pi a}{f}\right)\right)\right|^2\sim 0.52545\ldots.$$

Let u_1,u_2 form a fundamental system of units of K. Then, as $\mu_K=\{\pm 1\}$, $\epsilon_1=\pm u_1^a u_2^b$, $\epsilon_2=\pm u_1^c u_2^d$, for $a,b,c,d\in\mathbb{Z}$. Then,

$$\left|\det\begin{pmatrix}\log(|\sigma_1(\epsilon_1)|) & \log(|\sigma_1(\epsilon_2)|)\\ \log(|\sigma_2(\epsilon_1)|) & \log(|\sigma_2(\epsilon_2)|)\end{pmatrix}\right|=\left|\det\begin{pmatrix}\log(|\sigma_1(u_1^a u_2^b)|) & \log(|\sigma_1(u_1^c u_2^d)|)\\ \log(|\sigma_2(u_1^a u_2^b)|) & \log(|\sigma_2(u_1^c u_2^d)|)\end{pmatrix}\right|$$

$$=\left|\det\begin{pmatrix}a\log(|\sigma_1(u_1)|)+b\log(|\sigma_1(u_2)|) & c\log(|\sigma_1(u_1)|)+d\log(|\sigma_1(u_2)|)\\ a\log(|\sigma_2(u_1)|)+b\log(|\sigma_2(u_2)|) & c\log(|\sigma_2(u_1)|)+d\log(|\sigma_2(u_2)|)\end{pmatrix}\right|$$

$$=\left|\det\begin{pmatrix}\log(|\sigma_1(u_1)|) & \log(|\sigma_1(u_2)|)\\ \log(|\sigma_2(u_1)|) & \log(|\sigma_2(u_2)|)\end{pmatrix}\right|\cdot\left|\det\begin{pmatrix}a & c\\ b & d\end{pmatrix}\right|=|ad-bc|R_K.$$

Therfore, $|ad-bc|\sim 1.000\ldots$. As $ad-bc\in\mathbb{Z}$, it follows that rigorously $|ad-bc|=1$, i.e. $\begin{pmatrix}a & c\\ b & d\end{pmatrix}\in\mathrm{GL}_2(\mathbb{Z})$. This implies that we can take ϵ_1,ϵ_2 as a fundamental system of units of $\mathcal{O}_K^\times$, and therefore $\mathcal{O}_K^\times=C^+$.

$\square$